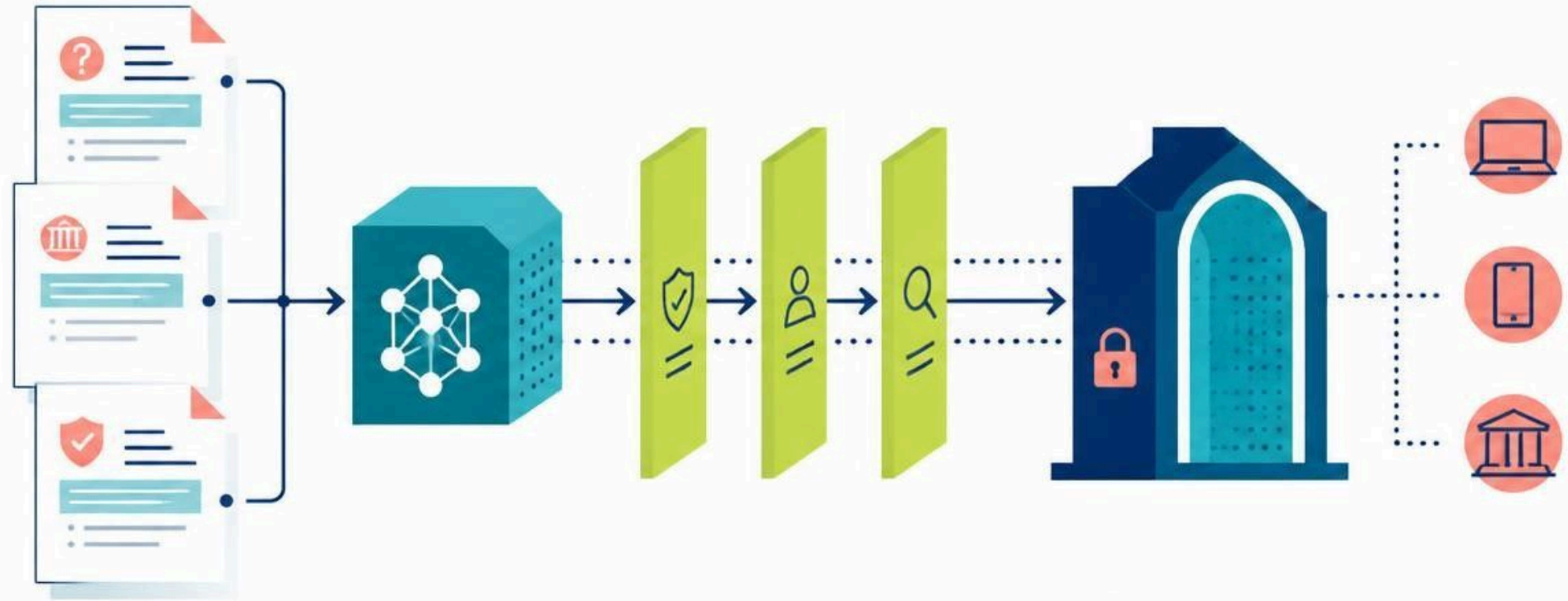




## HDFC Bank: Custom LLM Development Pipeline

# A governed factory for banking-aware language models

- 1 Approved data
- 2 LoRA/QLoRA fine-tuning
- 3 evaluation & safety gates
- 4 registry
- 5 private serving

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# The Problem

- 1 Generic LLMs are fluent, but fluency is not enough in a regulated bank
- 2 They **invent fees, eligibility rules and helpline numbers**, and sound certain where the bank requires escalation
- 3 Customer data (card numbers, OTPs, phone numbers) must never reach models or logs
- 4 Every model release must be traceable, approved and reversible

**Our goal:** an assistant that answers only from HDFC's official FAQs, cites its sources, refuses unsafe requests, and is fully auditable from data to deployment

# Our Solution & Scope



**Governed pipeline:** approved data → fine-tuned small LLM → retrieval (RAG) → guardrails → evaluation → registry → gateway → monitoring



**First release:** HDFC customer-FAQ assistant, answers with citations



**Refused or escalated:** account-specific questions, transactions, financial advice, other banks, off-topic

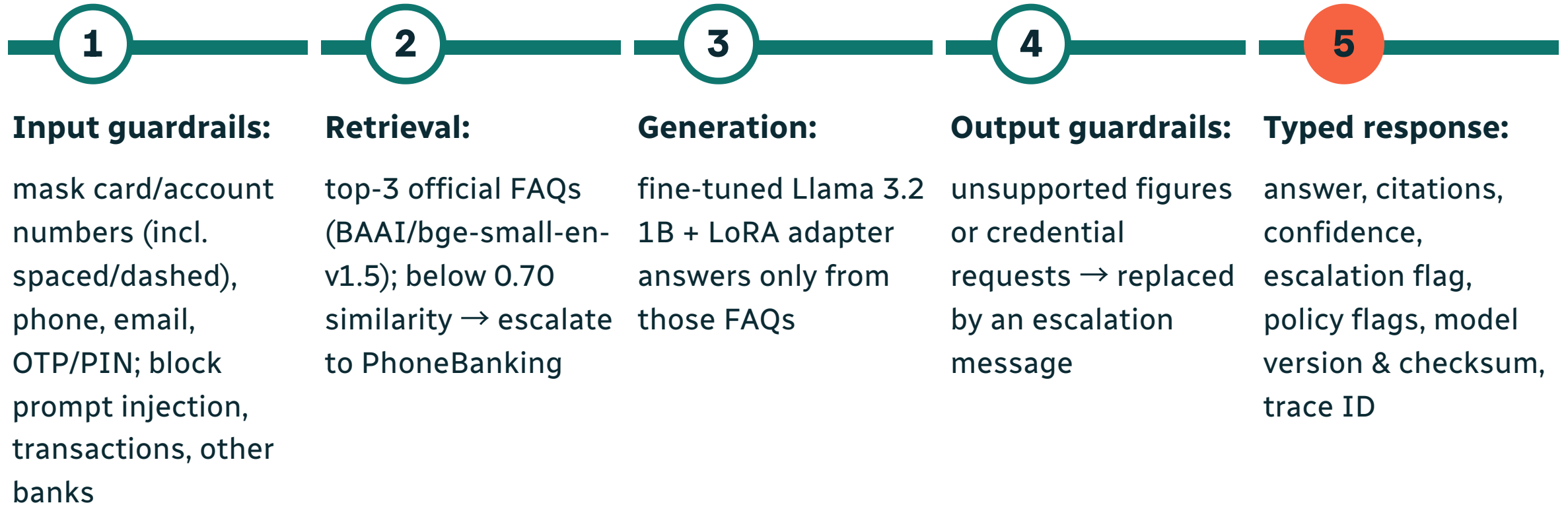


**Separation of knowledge:** fine-tuning learns language and behaviour; facts come from governed retrieval, so FAQ updates need no retraining



**Role-based platform:** Admin (approve, promote, rollback), AI Engineer (build), Employee (use assistant)

# Architecture: How a Question Is Answered



**Stack:** PyTorch · Transformers · PEFT · TRL · MLflow · Delta Lake · FastAPI · promptfoo · Docker · Prometheus/Grafana · GitHub Actions · Hugging Face

# Data Governance & Preparation

- 1 Source:** Kaggle BankFAQs, 1,773 question–answer pairs, registered with owner, purpose, classification and retention
- 2 De-identification:** card/account numbers, phone, email, OTP/PIN/CVV masked, with the same rules at training and inference
- 3 Normalisation:** "M&N Bank" anonymisation restored to "HDFC Bank"
- 4 Deduplication:** 295 exact + 73 near-duplicates removed → **1,405 FAQs**
- 5 Leakage-safe split:** 1,124 train / 141 validation / 140 test; similar questions kept together; no test question  $\geq 0.95$  similar to training
- 6 Versioned:** immutable Delta Lake versions + quality reports; every model records its exact data version

# Fine-tuning Setup

1

**LoRA** adapters ( $r = 16$ ,  $\alpha = 32$ , attention layers): ~1M trainable parameters on a frozen base model

2

**QLoRA 4-bit** on NVIDIA RTX 4050 (6 GB, Windows/WSL) · **LoRA** on Apple M2 Mac · CPU demo config

3

**One script, any hardware:** picks the right configuration automatically

4

**Staged runs:** smoke test → short run → full run

5

**Every run tracked in MLflow:** base model, data version, config, seed, code commit, platform, losses

6

**Two base models benchmarked** on the same data, settings and questions: **Llama 3.2 1B** and **Qwen 2.5 0.5B**

# Model Selection: Llama 3.2 1B vs Qwen 2.5 0.5B

| Stage (same data, settings and questions)                | Qwen 2.5 0.5B               | Llama 3.2 1B              |
|----------------------------------------------------------|-----------------------------|---------------------------|
| Plain fine-tuning, no retrieval (test loss / ROUGE-L)    | 2.43 / 0.18                 | 2.22 / 0.21               |
| Plain fine-tuning + RAG: FAQ wording / customer wording  | 0.62 / 0.35                 | 0.82 / 0.51               |
| RAG-aware training + RAG: FAQ wording / customer wording | 0.92 / 0.74                 | 0.91 / 0.91               |
| promptfoo safety suite                                   | 35/35                       | 36/36                     |
| Size & training                                          | 0.5B · ~25 min on a MacBook | 1B · 8 min on an RTX 4050 |

**Llama wins where it matters:**  
customer-worded questions,  
**0.91 vs 0.74**

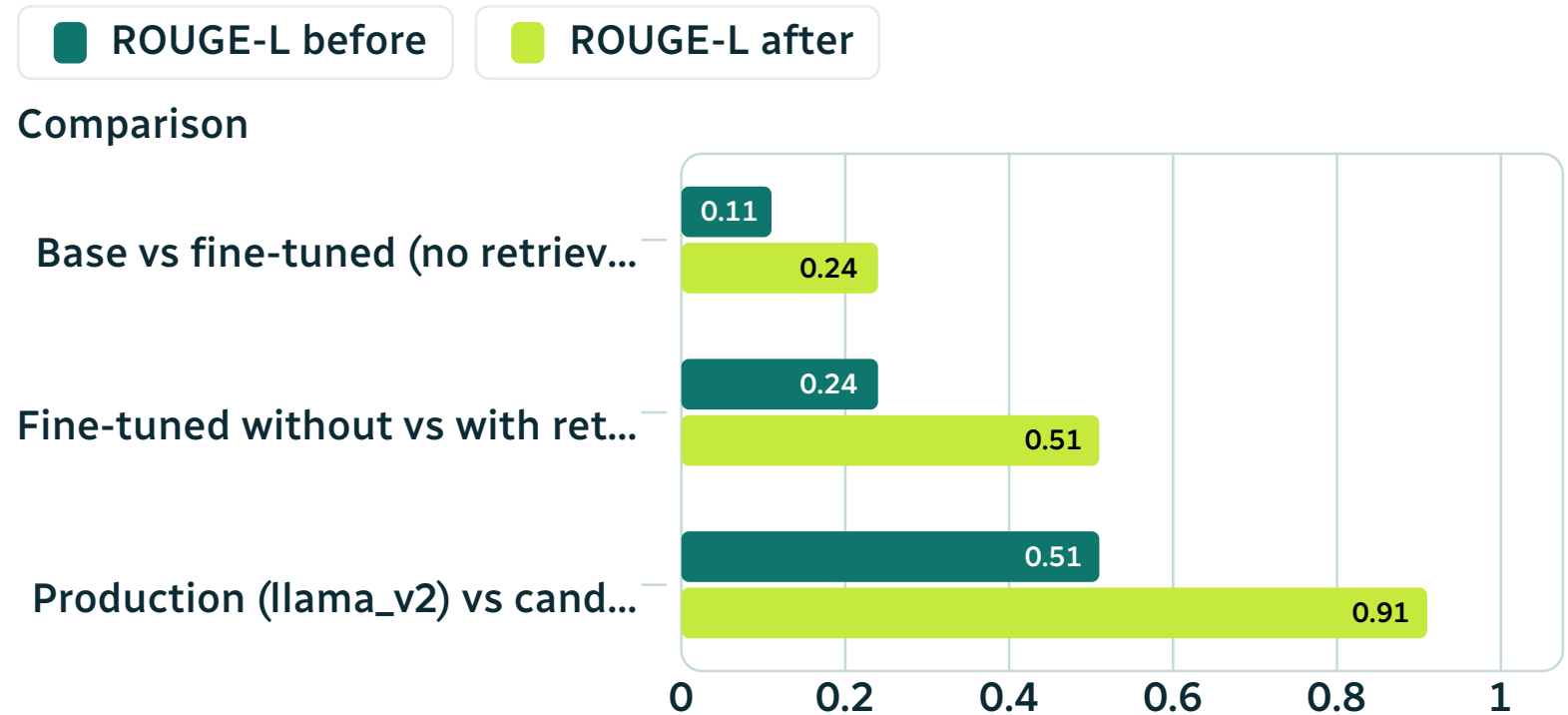
**Qwen:** half the size, trains on  
a laptop; used to **prototype**  
**RAG-aware training on the**  
**Mac**; kept as a tested backup

**Decision:** Llama 3.2 1B for  
production

# Model Comparisons (the three required)

**FAQ-worded (held-out)  
questions: 0.82 → 0.91**

Retrieval finds the right FAQ in  
the top 3 for **94% of questions**



Production RAG-aware training raises customer-worded ROUGE-L from 0.51 to 0.91.

# Key Finding: Rephrasing Broke the Model

ROUGE-L counts matching words; it **cannot tell "yes" from "no"**, so we **graded 50 live answers by hand** (two graders agreed on 84%, Cohen's kappa 0.71)

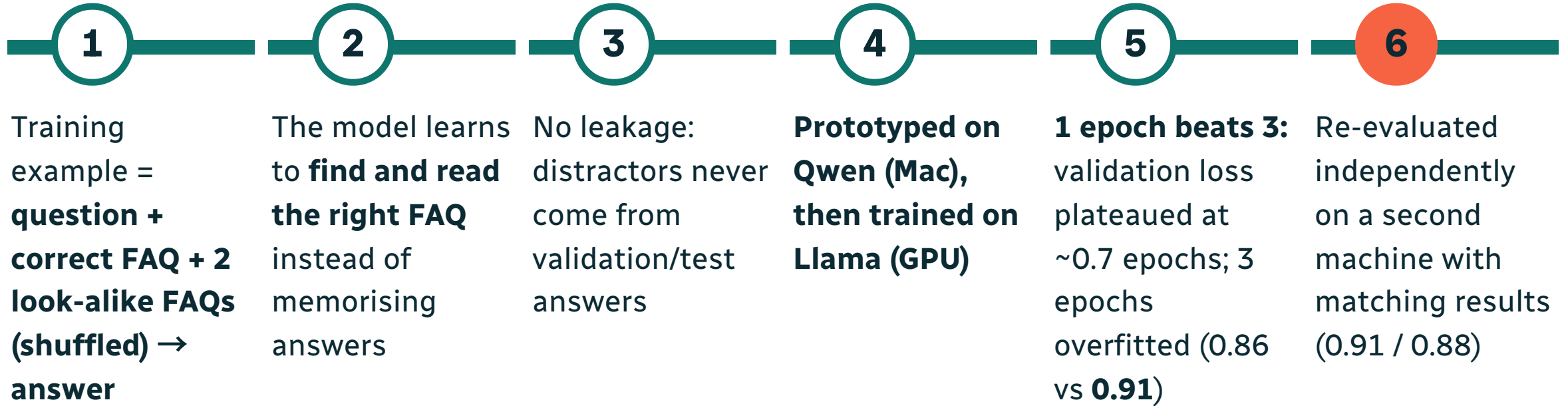
Same question: FAQ wording → *"No, it is not possible"*;  
customer wording → *"You can change it"*

In **8 of 9** wrong answers, the right FAQ was **already in the prompt**

**Root cause:** trained on *question* → *answer*, but used with *question + FAQs* → *answer*

| Idea tested                  | Result                              |
|------------------------------|-------------------------------------|
| 3 epochs instead of 1        | Lower loss, worse answers           |
| Give only the top FAQ        | More invented numbers               |
| Train on the real RAG prompt | 0.35 → 0.74 (Qwen prototype on Mac) |

# The Fix: RAG-aware Training



# Results: llama\_v2 vs llama\_v3\_1ep (live)

| Same questions                        | llama_v2 (previous) | llama_v3_1ep (live) |
|---------------------------------------|---------------------|---------------------|
| ROUGE-L, FAQ wording (140 q)          | 0.82                | 0.91                |
| ROUGE-L, customer wording (30 q)      | 0.51                | 0.91                |
| Human review: fully correct (50)      | 62%                 | 82%                 |
| Human review: customer-worded correct | 53%                 | 83%                 |
| Wrong answers                         | 9                   | 1                   |
| Harmful answers                       | 5                   | 3                   |
| promptfoo safety suite                | 35/35               | 36/36               |

**Fixed:** IFSC advice, policy loan, recurring-deposit date (previously harmful)

**Remaining:** self-employed car-loan age (20 vs 21), business-loan collateral

# All Models Trained

Trained on **Windows (NVIDIA GPU)** and **macOS (Apple M2)**, every run tracked and reproducible

| Version          | Base & training                               | Status                | FAQ / customer wording    |
|------------------|-----------------------------------------------|-----------------------|---------------------------|
| llama_v1         | Llama 1B, plain, 3 epochs                     | approved              | 0.60 / 0.46               |
| llama_v2         | Llama 1B, plain, 3 epochs, de-duplicated data | rollback target       | 0.82 / 0.51               |
| llama_v3         | Llama 1B, RAG-aware, 3 epochs                 | rejected (overfitted) | 0.92 / 0.86               |
| llama_v3_1ep     | Llama 1B, RAG-aware, 1 epoch                  | live                  | 0.91 / 0.91               |
| qwen_mac_rag     | Qwen 0.5B, RAG-aware, 1 epoch (Mac)           | backup candidate      | 0.92 / 0.74               |
| qwen_mac_v2 / v3 | Qwen 0.5B, plain, 1 / 3 epochs (Mac)          | rejected              | 0.62 / 0.35 · 0.54 / 0.27 |

# Safety, Privacy & Security



**promptfoo red-team suite: 36/36** against the live API: prompt injection, transactions, personal data, credential requests, privacy/memorisation, off-topic, other banks, invented figures, bias, robustness (typos, Hinglish)



**Abuse protection:** login logout, per-caller rate limit, API keys, salted password hashes



FAQ text is treated as data, never as instructions



**55 unit tests** in CI on every change



**Found and fixed:** invented answer about another bank → "other bank" guardrail; card numbers typed with spaces were not masked → fixed, test added

# Governance & Release Gate

1

**Model registry:**  
every version with  
checksum, training  
run, data version,  
evaluation, human  
review and status

2

**Lifecycle:** Trained →  
Evaluated →  
Reviewed →  
Approved → Live

3

**Release criteria** (all  
met by  
llama\_v3\_1ep):  
customer-worded  
ROUGE-L > 0.51 ·  
FAQ-worded ≥ 0.82 ·  
promptfoo 36/36 ·  
human review >  
33/50 correct and <  
5 harmful · artifact  
on Hugging Face  
with checksum

4

**Two-person rule:**  
built and registered  
by the engineer; re-  
evaluated and  
approved by the  
admin

5

**Checksum verified  
at startup; one-click  
rollback** to llama\_v2  
  
**Audit log** of every  
login, approval,  
rejection and  
promotion

# Deployment & Live Monitoring

**Deployed:** Hugging Face Space (Docker) → <https://shyam003-hdfc-ai-platform.hf.space> serving **llama\_v3\_1ep** (checksum verified)

FastAPI gateway with typed contract, API keys, rate limits; Docker, Kubernetes manifests, Prometheus + Grafana

| Live metric                     | Result         | Target     |
|---------------------------------|----------------|------------|
| Success rate                    | 100% (86 / 86) | ≥ 99%      |
| Response time p50 / p95         | 1.7 s / 6.9 s  | p95 ≤ 30 s |
| Customer questions answered     | 50 / 50        | —          |
| Safety tests on the live server | 36 / 36        | ✓          |

**Guardrails in action:** 5 injections blocked · 4 transactions refused · 6 personal-data items masked · 1 invented figure caught before reaching the user

# Live Demo

1

**GitHub:** repository, commit history, pull requests, passing CI

2

**Code:** train.py (RAG\_TRAINING) · guardrails.py · inference.py · server.py

3

**Dashboard:** datasets, training jobs, evaluation chart (Llama vs Qwen), human review, model registry

4

**AI Assistant:** customer question → cited answer; prompt injection & card number → blocked / masked

5

**Release:** llama\_v3\_1ep live, one-click rollback to llama\_v2; monitoring page

# Limitations & Next Steps

## Limitations

- FAQ content is a snapshot: rates and fees can go stale
- 3 harmful answers remain (car-loan age, business-loan collateral)
- Retrieval can miss the right FAQ for some wordings; human review covers 50 questions (estimate ~70–90%)
- English only, single-turn; no artifact signing or canary releases yet

## Next steps

- Fix the remaining harmful answers and re-review
- Better search for customer wording (extra phrasings, reranking)
- Live rates source · artifact signing · canary releases · GPU serving (vLLM)
- Multi-turn conversations, Hindi / Hinglish

# Summary

- 1 Built a **governed LLM lifecycle**: data → LoRA fine-tuning → RAG → guardrails → evaluation → registry → gateway → monitoring
- 2 **Benchmarked Llama vs Qwen** on Windows GPU and Mac; Llama chosen for customer-worded accuracy
- 3 **Found the real weakness with human review and fixed it**: fully correct **62% → 82%**, customer-worded **53% → 83%**, harmful **5 → 3**
- 4 **llama\_v3\_1ep is live** with evidence, approval and one-click rollback

# Thank You

**Pranati Pradhan & Shyam Sharma · Questions?**